Automatic Category Suggestion for Wikimedia Commons

Problem statement analysis, technical approach and evaluation rubric

1 Problem definition

Let $\mathcal{C}$ be the set of Commons categories, with $|\mathcal{C}|$ in the millions, organized as a directed graph $G = (\mathcal{C}, E)$ (a DAG in intent; cycles exist in practice). Given a newly uploaded image x and optional metadata m (title, description, EXIF, geotag, uploader), produce a ranked list of categories

$$\hat{Y}_k(x, m) = \text{top-}k_{c \in \mathcal{C}}\ s(c \mid x, m),$$

to be compared with the editor-assigned set $Y(x) \subseteq \mathcal{C}$. This is *extreme multi-label classification* with a hierarchical, long-tailed, noisily labelled and open-ended label space. The output is a suggestion for human review, not an automatic edit.

Requirements stated in the problem and what each implies

Requirement	Technical implication
Cover objects, scenes, events, abstract concepts	Heterogeneous label semantics. Visual features suffice for objects; events and abstract concepts need text and metadata (date, place, caption) as additional signal.
Large, varied labelled data	Labels are volunteer-assigned, so incomplete and inconsistent. Treat unlabelled as *unknown*, not negative.
Accuracy on *new* images	Temporal split and near-duplicate removal; a random split leaks.
Efficient, scalable, daily volume	Inference must be a bounded-cost forward pass plus an approximate nearest-neighbour (ANN) lookup; no per-category model.
Varied quality and resolution	Fixed-size resize at inference; train with blur, JPEG, downscale and crop augmentation.
Learn from new data	New categories must be addable without retraining the encoder; accept/reject feedback is a training signal.
Open, transparent, inclusive	Open weights and code, explainable suggestions (show neighbours), audit by region and language.

2 Data

Source. Commons database dumps and the MediaWiki API (`categorylinks`, `page`, image metadata). Restrict to files under open licences.

Label cleaning.

- Drop maintenance and tracking categories (licence templates, "Uploaded with…", backlog and cleanup categories) and hidden categories that encode process, not content.
- Resolve redirects and merge renamed categories.
- Keep categories with at least $n_{\min}$ training images (e.g. 20–50) for the supervised head. Rarer categories are served by retrieval and by text embeddings (Section 3).

Label structure. The category frequency distribution is heavy-tailed. Most files carry few categories, and editors file under the most specific applicable category rather than all ancestors, so the ancestor set $\text{anc}(c)$ is systematically *unlabelled* though correct.

Splits. Split by upload timestamp (train earlier, test later). Remove near-duplicates across splits by perceptual hash or embedding cosine similarity above a threshold. Group by uploader to prevent style leakage.

3 Model

The design uses a pre-trained vision–language encoder f_θ (CLIP/SigLIP-style or ViT) producing ℓ_2-normalized embeddings $z_x = f_\theta(x)$, and treats categories as retrievable objects rather than a fixed softmax output.

Component scorers

1. Instance retrieval (k-NN). Index embeddings of categorized images. For a query, take its K nearest neighbours $\mathcal{N}_K(x)$ and vote:

$$s_{\text{knn}}(c \mid x) = \frac{\sum_{i \in \mathcal{N}_K(x)} \exp(z_x^\top z_i / \tau)\, \mathbb{1}[c \in Y_i]}{\sum_{i \in \mathcal{N}_K(x)} \exp(z_x^\top z_i / \tau)}.$$

Needs no retraining when images or categories are added, and the neighbours serve as the explanation shown to users.

2. Text-prototype (zero-shot). Embed each category name (and aliases, multilingual labels) with the text tower: $s_{\text{txt}}(c \mid x) = z_x^\top t_c$. Covers categories with no or few examples.

3. Supervised multi-label head. A linear or small-MLP head on z_x (optionally with the last encoder blocks unfrozen) with a per-category sigmoid, $p_c = \sigma(w_c^\top z_x + b_c)$. With positives rare, use asymmetric loss

$$\mathcal{L} = -\sum_c \Big[\, y_c\,(1-p_c)^{\gamma_+} \log p_c + (1-y_c)\, p_{c,m}^{\gamma_-} \log(1-p_{c,m}) \Big], \quad p_{c,m} = \max(p_c - m, 0),$$

which down-weights easy negatives and tolerates missing labels through the margin m. For a very large $|\mathcal{C}|$, share parameters by initializing w_c from t_c (a text-initialized classifier).

4. Metadata scorer. A text model over title and description, plus geotag and date features, for events and places. Used as an additional input to the fusion step, not alone.

Fusion and hierarchy

Combine scorers by calibrated stacking, $s(c \mid x) = \sigma(\mathbf{w}^\top [s_{\text{knn}}, s_{\text{cls}}, s_{\text{txt}}, s_{\text{meta}}] + b)$, fit on a held-out validation split by logistic regression on candidate categories. Fusion operates only on the union of each scorer's top candidates (a few hundred), not all of $\mathcal{C}$.

Post-process with the graph: if a category is selected, suppress its ancestors to follow the most-specific convention; if an ancestor scores high but no descendant does, keep the ancestor. This directly addresses the unlabelled-ancestor bias noted above.

Robustness to quality

Train and test-time resize to the encoder's fixed input size. Augment with Gaussian blur, random down-scale/upscale, JPEG re-encoding and random-resized crop. Report metrics stratified by resolution band.

Continual learning

- **Index updates:** append embeddings of newly accepted images to the ANN index, with no retraining.
- **New categories:** add t_c and any examples to the index; they become available for s_{txt} and s_{knn} immediately. Extend the supervised head at the next scheduled retrain.
- **Feedback:** editor accept/reject events become labels, with rejects as hard negatives (these carry more reliable signal than absent labels).
- **Monitoring:** track acceptance rate and score calibration per category family; retrain on drift.

4 Serving and scale

Concern	Design and estimate
Throughput	Uploads are on the order of 10^4–10^5 per day (approximate), i.e. roughly 1 image/s on average at the upper end. A single GPU, or batched CPU inference with a quantized ViT-B encoder, covers this; compute is not the bottleneck.
Index size	For 10^8 images at 512-d fp16: $10^8 \times 512 \times 2$ B ≈ 102 GB flat. Product quantization at 64 B/vector gives ≈ 6.4 GB; HNSW or IVF-PQ queries run in milliseconds.
Pipeline	Upload event → queue → encoder → ANN lookup → fusion → store suggestions. Asynchronous, so it never blocks the upload.
Model cost	Distill or quantize the encoder (INT8, ONNX/TensorRT) after measuring accuracy loss on the held-out set.

5 Evaluation

The statement specifies "accuracy in suggesting the correct categories for new images." Single-label top-1 accuracy does not apply: $|Y(x)| > 1$ and the label space is huge. Use ranking metrics over the suggested list:

$$\mathrm{P@}k = \frac{|\hat{Y}_k \cap Y|}{k}, \qquad \mathrm{R@}k = \frac{|\hat{Y}_k \cap Y|}{|Y|}, \qquad \mathrm{F1@}k = \frac{2\,\mathrm{P@}k\,\mathrm{R@}k}{\mathrm{P@}k + \mathrm{R@}k}.$$

Average per image (sample-wise) for the user-facing quality, and macro-average per category for tail behaviour. Also report mean average precision (mAP) over categories and, as a hierarchy-aware variant, precision where a predicted ancestor or descendant of a true label counts as partially correct.

Protocol.

- Temporal hold-out with deduplication (Section 2). Report results for head, torso and tail categories separately.
- Stratify by category type (object, scene, event, abstract), image resolution and language of the category label.
- Because stored labels are incomplete, estimate false-negative inflation by human review of a sample of "wrong" suggestions. Report the strict score and the reviewed score.
- Report latency (p50/p95 per image), throughput, index size and memory.

6 Evaluation rubric

The problem statement defines no scoring rubric. The table below is *inferred* from the stated requirements; the weights are estimates, not the organizers' values.

Criterion	Weight	Evidence a judge can check
Accuracy on new images	40%	P@k, R@k, F1, mAP on a temporal hold-out; tail results included.
Category breadth	10%	Metrics by object / scene / event / abstract.
Efficiency and scalability	15%	Latency, throughput, index memory, serving design.
Robustness	10%	Metrics by resolution and degradation.
Continual learning	10%	Index update path, feedback loop, retraining policy.
Wikimedia fit and usability	10%	Open licences, explanations, human-in-the-loop UI, regional bias audit.
Presentation	5%	Clear demo, honest limitations.

7 Risks and mitigations

Risk	Mitigation
Train/test leakage from near-duplicate uploads	Temporal split, perceptual-hash and embedding deduplication, uploader grouping.
Missing labels treated as negatives	Asymmetric loss with margin; reviewed-sample evaluation; use reject events as true negatives.
Model learns process categories	Remove maintenance and hidden process categories before training.
Tail categories ignored by micro metrics	Macro metrics; retrieval and text-prototype scorers for rare labels.
Geographic and cultural bias	Stratified reporting by region and language; multilingual category names; sampling that does not follow upload volume only.
Sensitive content and people	No identification of individuals; suggestions limited to content categories; human approval required.